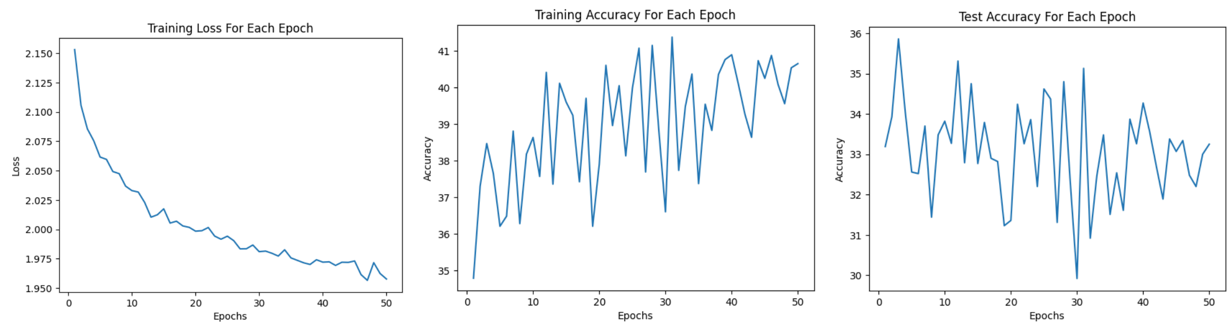
Zachary Foster

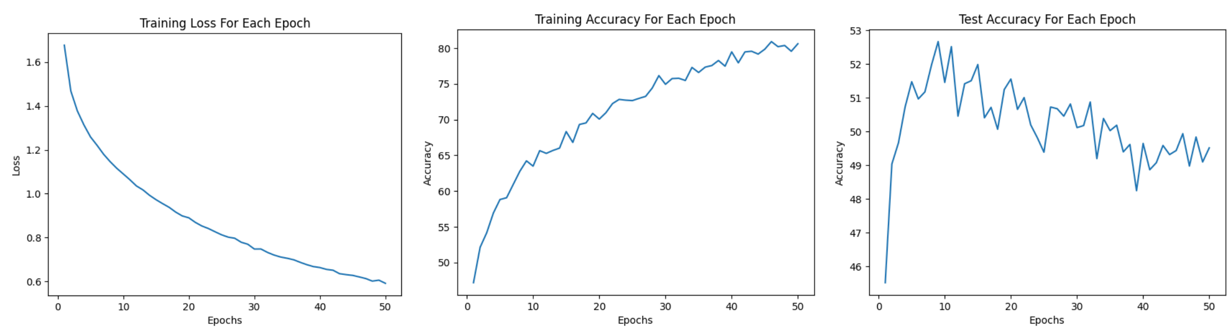
<u>Problem 2:</u> (50 epochs)
What is the number of parameters?

(Number of Pixels)*(Number of Weights) + (Bias of Outputs)
(32*32*3)*(10) + 10 = **30,730**

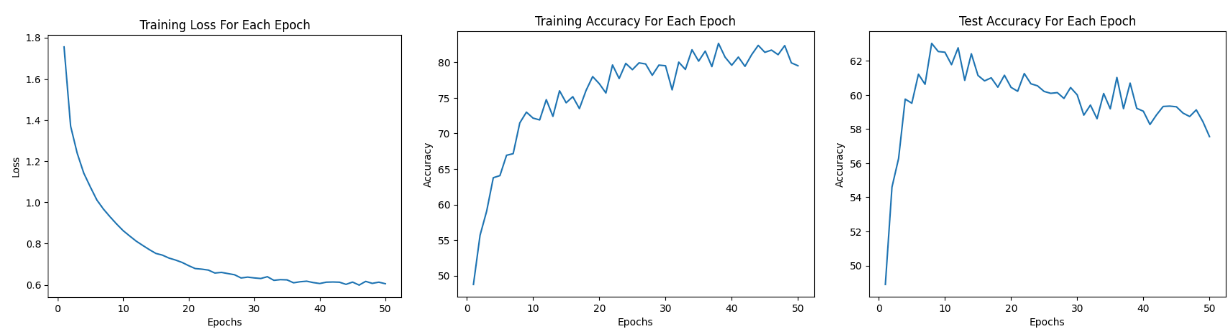


<u>Problem 3:</u> (50 epochs)
What happens if you do not use ReLU?

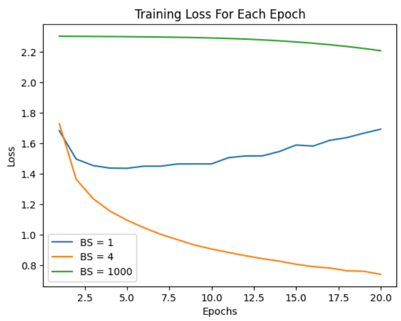
If you do not include a nonlinearity, then the entire network is effectively just one layer, which reduces its overall complexity and makes it more difficult to map nonlinear relationships in the data.

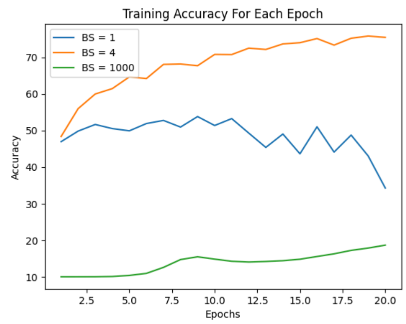


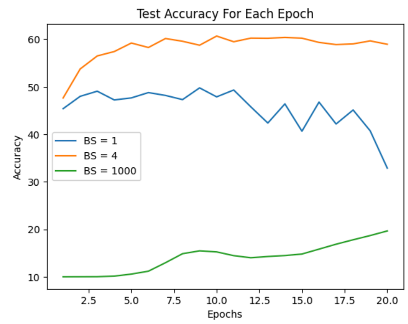
<u>Problem 4:</u> (50 epochs)
There is definitely a problem with overfitting after epoch 10 or so.



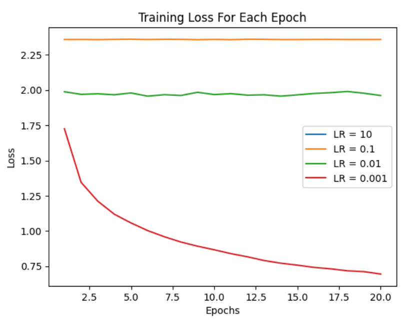
Problem 5: (20 epochs)
I had to reduce the number of epochs due to the amount of time it would take to train three models. A batch size of 1 took a very long time to train and didn't decrease loss as the epochs progressed. In contrast, a batch size of 1000 trained very quickly; however, the loss also reduced only slightly. The batch size of 4 seemed to be very similar to the training within problem 4.

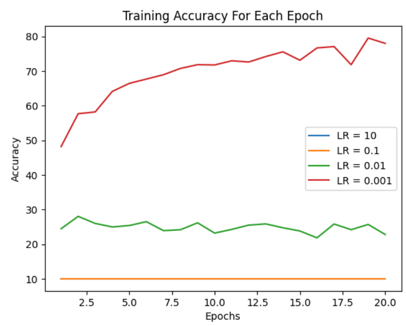


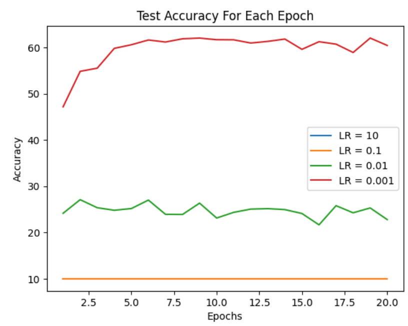




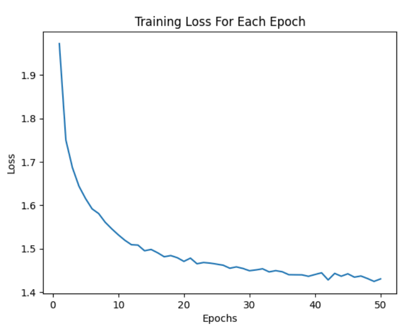
Problem 6: (20 epochs)
The LR 10 values are completely missing due to them being nan values; it seems like an LR of 0.001 is the only value that actually worked for the training.

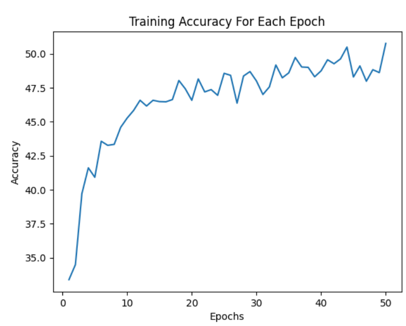


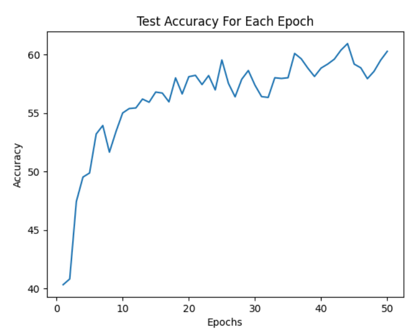




Problem 7: (50 epochs)
There was a notable reduction in overfitting, which is evident in that the test accuracy continued to increase through all of the epochs.







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Problem 8: (50 epochs)
The training was much smoother than it was with Cross-Entropy. Interestingly, it still achieved a test accuracy of about 65% and did not overtrain.